



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO LAXMI FUELS Indian Oil Dealer

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

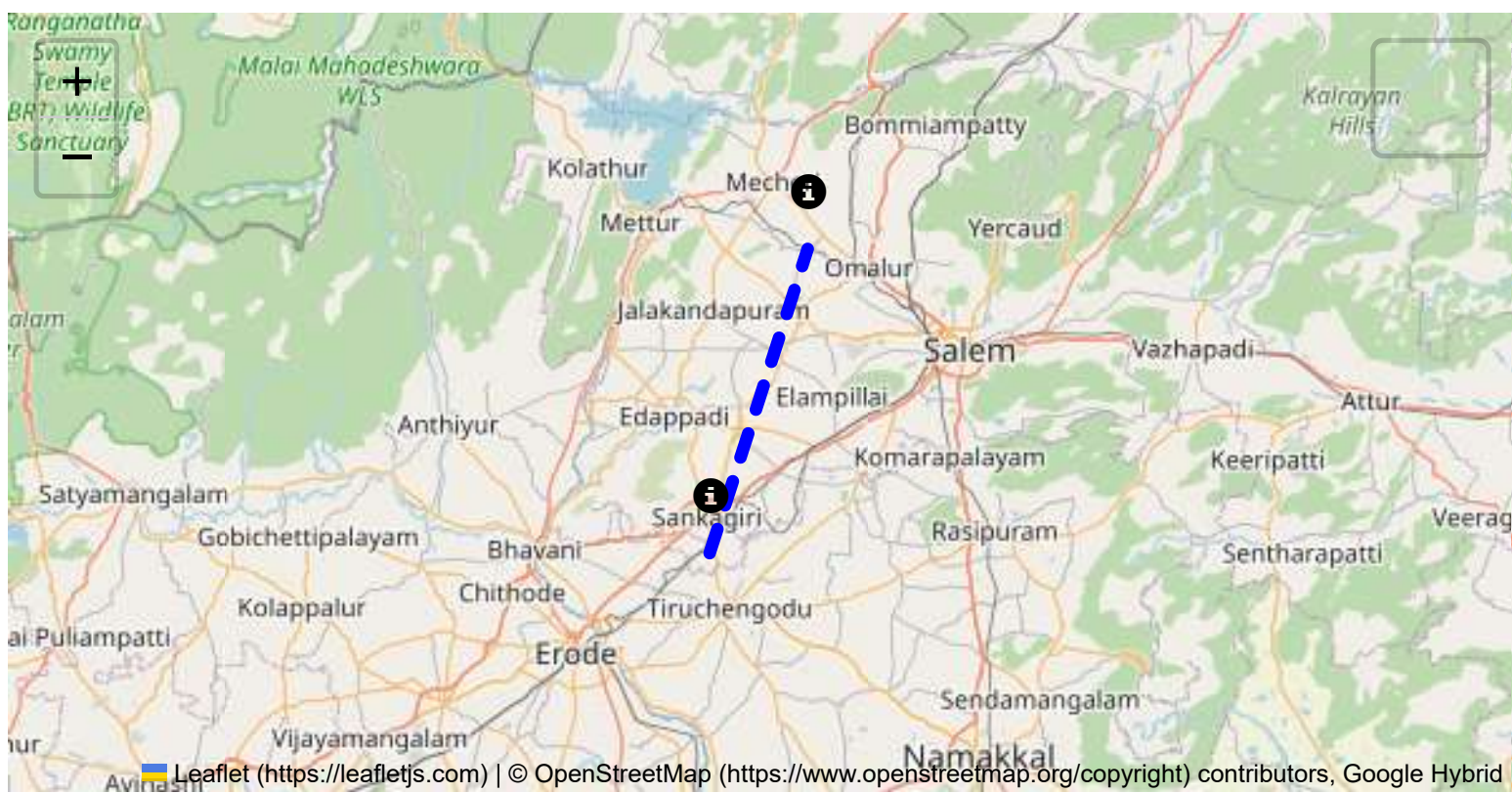
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

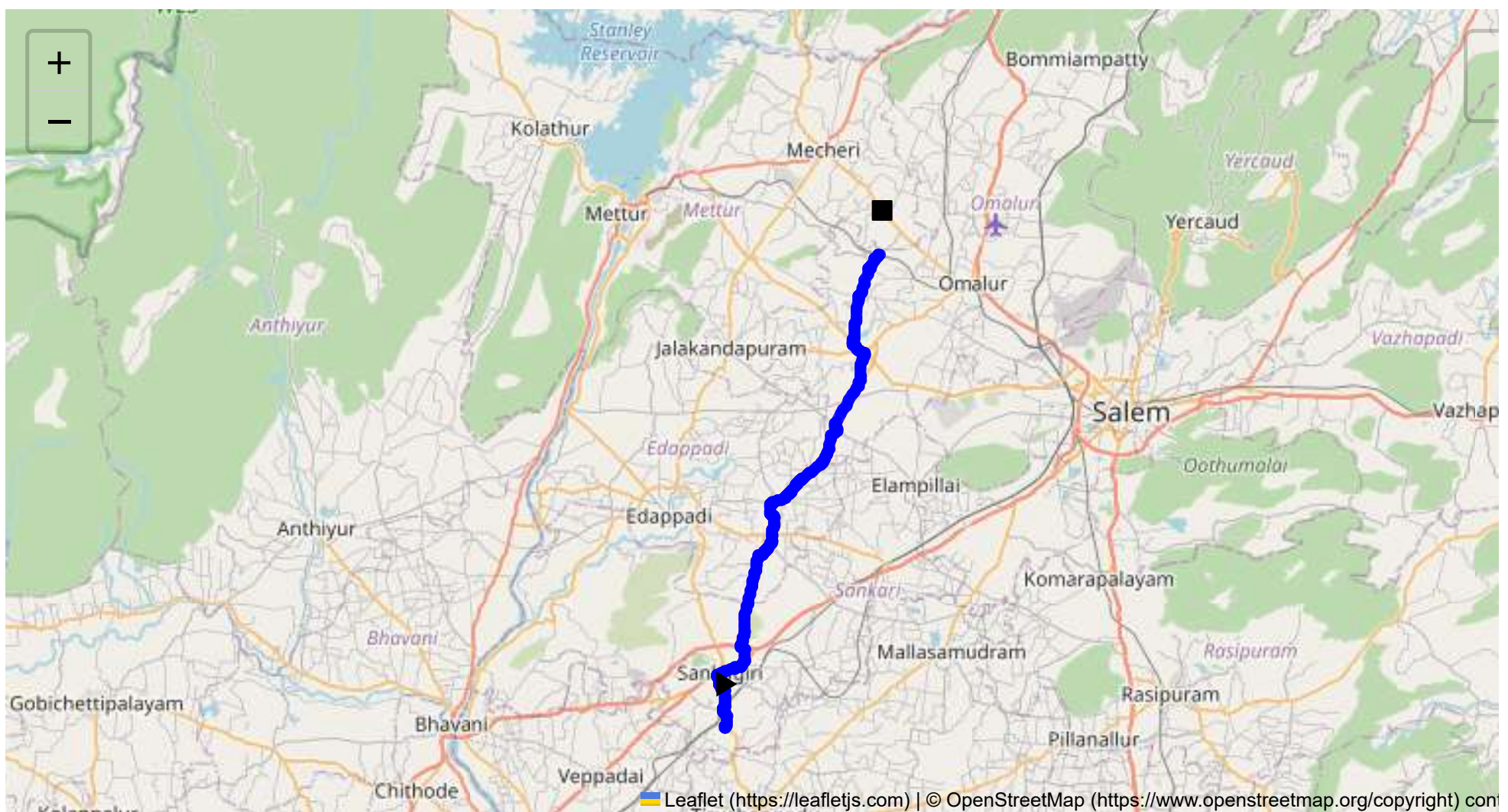
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 42.75 km
Estimated Duration: 1.2 hours
Adjusted Duration (Heavy Vehicle): 1.4 hours
Start: (11.4381, 77.8734)
End: (11.762612, 77.981918)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map: The route from Sangagiri to T.Maramangalam spans approximately 42.75 kilometers, primarily following rural roads with moderate traffic load. It traverses through small towns and agricultural areas, requiring navigation through a mix of state highways and local roads, with limited expressway sections.

2. Typical Weather Conditions and Potential Hazards: Tamil Nadu experiences a tropical climate with the southwest monsoon (June-September) and northeast monsoon (October-December) significantly affecting the weather. Heavy rains can cause road flooding and compromised visibility, particularly during monsoon seasons, creating potential hazards. Drivers should exercise caution during these months.

3. Analysis of Traffic Patterns: Traffic is generally moderate, but peaks during morning (7-9 a.m.) and evening (5-7 p.m.) times, especially around towns like Sankari and Edappadi, where local commuters contribute to congestion. These areas might have 2-wheeler traffic and pedestrians, which increase risk for heavy vehicles.

4. Assessment of Road Quality and Infrastructure: The road quality varies, with certain segments having well-maintained surfaces, while others might have potholes due to heavy rains. Narrow lanes, occasional speed breakers, and unmarked roads in some sections can challenge navigation for heavy vehicles. There are relatively few rest areas or facilities for trucks.

5. Suggestions for Alternative Routes for Emergencies: In case of road closures or emergencies, detouring via NH 544 towards Salem and then taking local roads might be viable but could add distance. Local knowledge of roads can offer detours through minor roads that bypass congestion.

6. Summary of Local Regulations Affecting Hazardous Material Transport: Tamil Nadu has specific regulations regarding the transportation of hazardous materials, which require special permits and adherence to designated transportation routes. Daytime travel might be preferred to avoid rural road hazards at night.

7. Overview of Historical Incidents: The region has seen minimal incidents involving hazardous materials, though heavy vehicle accidents have occurred due largely to road quality and driver errors. Preventive caution in known accident-prone areas, like bends or intersections in small towns, is advised.

8. Environmental Considerations and Sensitive Areas: The route passes through agricultural zones where spillage of hazardous materials can impact soil and water quality. These are environmentally sensitive areas, and any incidents would require prompt containment.

9. Analysis of Communication Coverage: Telecom coverage is generally good along major parts of the route but may experience dead zones in more rural or hilly areas. Satellite phones or alternative communication devices are recommended for emergencies.

10. Estimated Emergency Response Times: Response times can vary - urban areas can expect help within 30-45 minutes, while remote locations might see delays up to 1.5 hours. It's vital to have emergency contact numbers ready and awareness of the nearest medical facilities.

12. Overall Summary of Risk Assessment: The route features risks related to road quality, traffic congestion in towns, and variable weather conditions impacting travel safety. The main hazards include navigating narrow roads during peak hours, potential monsoon-related disruptions, and handling environmental sensitivities in agricultural zones. Efficient communication and knowledge of alternative paths are crucial for mitigating risks, alongside compliance with transport regulations for hazardous materials. Regular updates on traffic and weather conditions can significantly improve safety for heavy vehicle operators on this route.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	High	11.45004, 77.87206	15 KM/Hr
5	Turn	Medium	11.45028, 77.87202	30 KM/Hr
6	Turn	High	11.47403, 77.86841	15 KM/Hr
7	Turn	Medium	11.47410, 77.86845	30 KM/Hr
8	Turn	Medium	11.48363, 77.88780	30 KM/Hr
9	Turn	Medium	11.49451, 77.88489	30 KM/Hr
10	Turn	High	11.55614, 77.89764	15 KM/Hr
11	Turn	High	11.58317, 77.90647	15 KM/Hr
12	Turn	Medium	11.63840, 77.94820	30 KM/Hr
13	Turn	Medium	11.64733, 77.95122	30 KM/Hr
14	Blind Spot	Blind Spot	11.69590, 77.97114	10 KM/Hr
15	Turn	High	11.70129, 77.96360	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
1	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
2	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
3	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
4	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
6	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
8	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
9	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
10	hospital	Dr. Pandiyarajan Hospital	11.692704, 77.969344	30 km/h	Medium
11	clinic	Sun Eye Care	11.6936242, 77.9730948	30 km/h	Medium
12	hospital	Dr. Sengodan Hospital	11.6959527, 77.9695325	30 km/h	Medium
13	hospital	Veda Maruthuvamanai Hospital	11.6956155, 77.9678122	30 km/h	Medium
14	clinic	E.V.P Vairavel Clinic	11.6954975, 77.9667903	30 km/h	Medium
15	clinic	AMS Clinic	11.752664, 77.974974	30 km/h	Medium
16	hospital	Tholasampatti, Government Hospital	11.75638, 77.973813	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
7	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45004, 77.87206



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45028, 77.87202



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.47403, 77.86841



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.47410, 77.86845



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.48363, 77.88780



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49451, 77.88489



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.55614, 77.89764



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58317, 77.90647



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63840, 77.94820



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64733, 77.95122



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.69590, 77.97114



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.70129, 77.96360
